

DAFTAR SINGKATAN DAN LAMBANG

SINGKATAN	Nama	Hal.
AI	<i>Artificial Intelligence</i>	2
ANRI	<i>Arsip Nasional Republik Indonesia</i>	1
API	<i>Application Programming Interface</i>	85
BCE	<i>Binary Cross-Entropy</i>	30
BiGRU	<i>Bidirectional Gated Recurrent Unit</i>	25
BiLSTM	<i>Bidirectional Long Short-Term Memory</i>	25
CBAM	<i>Convolutional Block Attention Module</i>	35
CER	<i>Character Error Rate</i>	5
cGAN	<i>Conditional Generative Adversarial Network</i>	20
CLI	<i>Command-Line Interface</i>	90
CNN	<i>Convolutional Neural Network</i>	15
CRNN	<i>Convolutional Recurrent Neural Network</i>	20
CTC	<i>Connectionist Temporal Classification</i>	25
CUDA	<i>Compute Unified Device Architecture</i>	90
cuDNN	<i>CUDA Deep Neural Network library</i>	90
CWV	<i>Core Web Vitals</i>	95
DE-GAN	<i>Document Enhancement GAN</i>	18
DIBCO	<i>Document Image Binarization Contest</i>	22
DocEnTr	<i>Document Enhancement Transformer</i>	19
DRD	<i>Distance Reciprocal Distortion</i>	22
DSRM	<i>Design Science Research Methodology</i>	50
ERB	<i>Enhance to Read Better</i>	18
FCN	<i>Fully Convolutional Network</i>	16
FM	<i>F-measure</i>	22
Fps	<i>pseudo-F-measure</i>	22
FR	<i>Functional Requirement</i>	60
GAN	<i>Generative Adversarial Network</i>	2
GPU	<i>Graphics Processing Unit</i>	85
GT	<i>Ground Truth (Kebenaran Dasar)</i>	55
H-DIBCO	<i>Handwritten Document Image Binarization Contest</i>	22
HMM	<i>Hidden Markov Model</i>	25
HTR	<i>Handwritten Text Recognition</i>	3
IAM	<i>IAM Handwriting Database</i>	27
ICDAR	<i>International Conference on Document Analysis and Recognition</i>	22
IIIT5K	<i>IIIT 5K-word dataset</i>	27
KHATT	<i>Arabic Handwritten Text Database</i>	27
LSTM	<i>Long Short-Term Memory</i>	25
MAE	<i>Mean Absolute Error</i>	30

ML	<i>Machine Learning</i>	2
MLflow	<i>Machine Learning Flow</i>	90
MSE	<i>Mean Squared Error</i>	30
MSFP	<i>Multi-Scale Feature Pyramid</i>	35
NFR	<i>Non-Functional Requirement</i>	65
OCR	<i>Optical Character Recognition</i>	3
PALM	<i>Presentation Attack Liveness Detection in Mobile scenarios</i>	27
Pix2Pix	<i>Image-to-Image Translation with Conditional Adversarial Networks</i>	19
PSNR	<i>Peak Signal-to-Noise Ratio</i>	5
RDB	<i>Residual Dense Block</i>	35
RMSProp	<i>Root Mean Square Propagation</i>	85
RNN	<i>Recurrent Neural Network</i>	25
SDM	<i>Sumber Daya Manusia</i>	60
SGD	<i>Stochastic Gradient Descent</i>	85
SNR	<i>Signal-to-Noise Ratio</i>	30
SOP	<i>Standard Operating Procedure</i>	65
SOTA	<i>State-of-the-Art</i>	5
SSIM	<i>Structural Similarity Index Measure</i>	5
UNESCO	<i>United Nations Educational, Scientific and Cultural Organization</i>	1
U-Net	<i>U-shaped Convolutional Network</i>	16
VGG	<i>Visual Geometry Group</i>	17
ViT	<i>Vision Transformer</i>	19
VOC	<i>Visual Object Classes</i>	27
WER	<i>Word Error Rate</i>	28

LAMBANG	NAMA	Hal.
α	Koefisien bobot untuk <i>loss function</i> , tingkat signifikansi statistik	75
β	Koefisien bobot untuk <i>Binary Cross-Entropy loss</i>	75
λ	Koefisien bobot untuk <i>CTC loss</i> , parameter regularisasi	75
θ	Parameter model <i>neural network</i>	75
σ	Deviasi standar	75
μ	Rata-rata (<i>mean</i>)	75
ε	<i>Token blank</i> /kosong dalam <i>CTC</i> , konstanta kecil untuk stabilitas numerik	75
Δ	Delta (selisih/perubahan nilai)	75
π	<i>Alignment path</i> dalam <i>CTC</i>	75
π_t	<i>Token</i> pada <i>timestep t</i> dalam <i>alignment path</i>	76
∇ atau ∂	Operator gradien atau turunan parsial	75
Σ	Operator penjumlahan	75
Π	Operator perkalian	75
$\int$	Operator integral	75
argmax	Argumen yang memaksimalkan fungsi	76
$\sim$	Terdistribusi menurut	75
$\rightarrow$	Menuju atau konvergen ke	75
$\mathcal{L}$	Fungsi <i>loss</i> (kerugian)	75
$\mathcal{L}_{total}$	Total <i>loss function</i>	75
$\mathcal{L}_{adversarial}$ atau $\mathcal{L}_{adv}$	<i>Adversarial loss</i>	75
$\mathcal{L}_{reconstruction}$	<i>Reconstruction loss</i>	75
$\mathcal{L}_{L1}$	<i>L1 loss (Mean Absolute Error)</i>	75
$\mathcal{L}_{L2}$	<i>L2 loss (Mean Squared Error)</i>	75
$\mathcal{L}_{CTC}$	<i>CTC (Connectionist Temporal Classification) loss</i>	75
$\mathcal{L}_{perceptual}$	<i>Perceptual loss</i>	75
$\mathcal{L}_{pixel}$	<i>Pixel-wise loss</i>	75
$\mathcal{L}_{BCE}$	<i>Binary Cross-Entropy loss</i>	75
$\mathcal{L}_{GAN}$	<i>GAN loss</i>	75
$\mathcal{L}_{cycle}$	<i>Cycle consistency loss</i>	75
G	<i>Generator</i> dalam GAN	75
D	<i>Discriminator</i> dalam GAN	75
$G(z)$	<i>Output</i> dari <i>generator</i> dengan <i>input z</i>	75
$D(x)$	<i>Output</i> dari <i>discriminator</i> dengan <i>input x</i>	75
$G_{A \rightarrow B}$	<i>Generator</i> dari domain <i>A</i> ke <i>B</i>	75
$G_{B \rightarrow A}$	<i>Generator</i> dari domain <i>B</i> ke <i>A</i>	75
$V(D, G)$	Fungsi nilai (<i>value function</i>) dalam GAN	75

x	Sampel data asli, <i>input</i> citra	75
y	Label atau <i>ground truth</i> , kondisi dalam <i>conditional GAN</i>	75
z	<i>Noise vector</i> atau representasi laten	
$\mathbb{E}$	Ekspektasi atau nilai harapan	75
p_{data}	Distribusi data asli	75
p_z	Distribusi <i>prior (noise)</i>	75
$p(y x)$	Probabilitas bersyarat <i>output y given input x</i>	75
$p_t(\pi_t x)$	Probabilitas <i>token</i> pada <i>timestep t</i>	
$\mathcal{B}$	Fungsi penciutan (<i>collapse function</i>) dalam CTC	75
$\mathcal{B}^{-1}$	Fungsi invers penciutan dalam CTC	75
T	Jumlah <i>timestep</i> atau panjang <i>sequence</i>	
$ x - G(y) _1$	L1 <i>distance</i> antara <i>ground truth</i> dan <i>generated image</i>	75
$ x - G(y) _2$	L2 <i>distance</i> antara <i>ground truth</i> dan <i>generated image</i>	75
$N \times N$	Ukuran <i>patch</i> dalam PatchGAN	
d	Cohen's d (<i>effect size</i>)	75
n	Jumlah sampel	75
p	p -value (nilai probabilitas statistik)	75
r	Koefisien korelasi	75
R^2	Koefisien determinasi	
$O(n)$	Notasi Big-O untuk kompleksitas linear	75
$O(n^2)$	Notasi Big-O untuk kompleksitas kuadratik	75